

Supplementary Note 1. Secondary selection-conditioned diagnostics

This note records diagnostics that are useful for scope control but are not used as source-wide estimates or as evidence for model behavior on the 43,139-structure Materials Project–Alexandria denominator.

Earlier 287-structure scored subset

ALIGNN-FF, CHGNet and MACE-MP raw scores were available only for an earlier 287-structure scored diagnostic subset. The three-model subset results are superseded for benchmark-impact claims by the 5,000-structure CHGNet sensitivity check in the main text. They remain useful only as a negative raw-score diagnostic: top-decile discordance was 5/29 for ALIGNN-FF, 3/29 for CHGNet and 3/29 for MACE-MP, and rank-permutation tests for the raw-score percentile coefficient were non-significant ($p=0.633$, 0.367 and 0.327 , respectively). These results should not be interpreted as model rankings.

WBM–Alexandria archived probe

An archived WBM–Alexandria probe is retained to explain why a previous high-discordance observation should not be read as a generic DFT-label discordance rate. The WBM probe gives 52.2% discordance on its own release-conditioned denominator (141/270), whereas the full MP–Alexandria MP-ID denominator gives 11.7%. Applying a simple MP-native exact-stable selection to the full MP–Alexandria denominator raises discordance to 21.5% (3,628/16,872), and an MP-native $e_{\text{hull}} \leq 25$ meV atom⁻¹ sensitivity set gives 17.1% (4,677/27,280). These comparisons support a limited conclusion: source pair, denominator and selection condition must be reported together.